

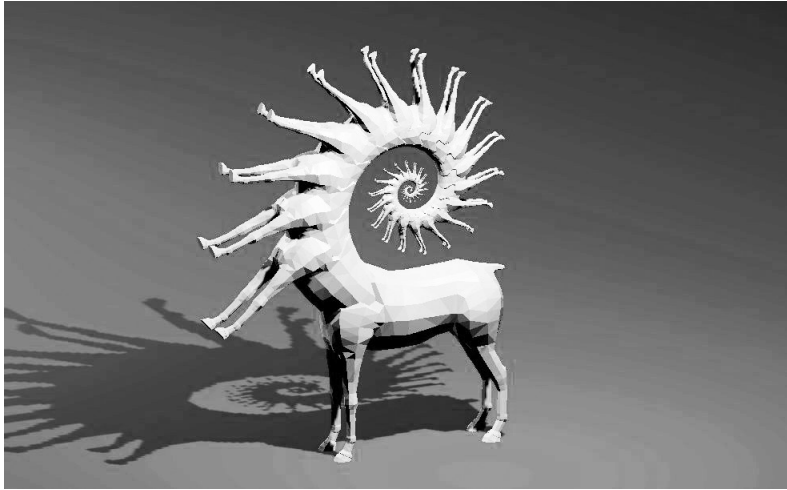


Figure 1-3: The recursive centaur. Image by Joseph Parker.

Based on these jokes, you might conclude that recursion is a sort of meta, self-referencing, dream-within-a-dream, infinite mirror-into-mirror sort of thing. Let's establish a concrete definition: a *recursive* thing is something whose definition includes itself. That is, it has a self-referential definition.

The Sierpiński triangle in Figure 1-4 is defined as an equilateral triangle with an upside-down triangle in the middle that forms three new equilateral triangles, each of which contains a Sierpiński triangle. The definition of Sierpiński triangles includes Sierpiński triangles.

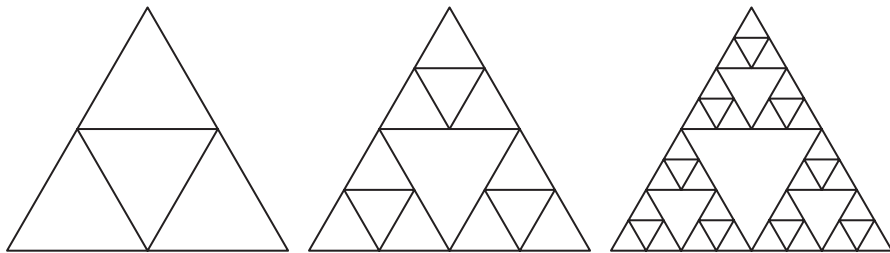


Figure 1-4: Sierpiński triangles are fractals (recursive shapes) that include Sierpiński triangles.

In a programming context, a *recursive function* is a function that calls itself. Before we explore recursive functions, let's take a step back and understand how regular functions work. Programmers tend to take function calls for granted, but even experienced programmers will find it worthwhile to review functions in the next section.

What Are Functions?

Functions can be described as mini-programs inside your program. They're a feature of nearly every programming language. If you need to run identical